

MATH 345 Skeleton Notes

Unit 4: Orthogonality and least squares

Section 4.3: Least squares as projection

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Revisiting a requested output as equations (U2-LO1, U2-LO3) asked whether a target vector $\mathbf{b}$ is reachable as $A\mathbf{x}$. When $\mathbf{b} \notin \text{col}(A)$, no input reaches the target exactly. Least squares replaces the unreachable target by the closest reachable output.

Recall the Unit 2 matrix and target

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

Every reachable output therefore has third coordinate equal to the sum of its first two coordinates. The target $\mathbf{b}$ is not reachable because $4 \neq 1 + 2$.

Definition: Least-squares solution, fitted vector, and residual

Let A be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$. A least-squares solution is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes

$$\|A\mathbf{x} - \mathbf{b}\|^2$$

over all $\mathbf{x} \in \mathbb{R}^n$.

The vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}$$

is the fitted vector. The vector

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \mathbf{b} - A\hat{\mathbf{x}}$$

is the residual.

The quantity

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^m ((A\mathbf{x})_i - b_i)^2$$

is the sum of the squares of the coordinate errors. This explains the name least squares. Minimizing the norm or its square gives the same minimizers.

If $\mathbf{b} \in \text{col}(A)$, the minimum error is 0, and least squares agrees with solving $A\mathbf{x} = \mathbf{b}$ exactly. If the target is not reachable, the fitted vector $\hat{\mathbf{b}}$ is the closest reachable output.

In the fig-u4-projection-residual, take the subspace to be $\text{col}(A)$, relabel the original vector as $\mathbf{b}$, and relabel its projection as

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}.$$

The residual is

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

The projection theorem guarantees a closest vector $\hat{\mathbf{b}} \in \text{col}(A)$. Since $\hat{\mathbf{b}}$ lies in the column space, at least one coefficient vector $\hat{\mathbf{x}}$ satisfies

$$A\hat{\mathbf{x}} = \hat{\mathbf{b}}.$$

Fact: A^T checks column orthogonality

Write

$$A = [\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_n].$$

For any vector $\mathbf{r} \in \mathbb{R}^m$,

$$A^T \mathbf{r} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{r} \\ \vdots \\ \mathbf{a}_n^T \mathbf{r} \end{bmatrix}.$$

Therefore

$$A^T \mathbf{r} = \mathbf{0}$$

exactly when $\mathbf{r}$ is orthogonal to every column of A . Equivalently,

$$\text{col}(A)^\perp = \text{null}(A^T).$$

Every fitted vector $A\hat{\mathbf{x}}$ lies in $\text{col}(A)$. Therefore the least-squares fitted vector is

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = \text{proj}_{\text{col}(A)}(\mathbf{b}).$$

Its residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$$

is orthogonal to $\text{col}(A)$.

Theorem: Least squares and the normal equations

A vector $\hat{\mathbf{x}}$ is a least-squares solution exactly when the residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$$

is orthogonal to $\text{col}(A)$. Equivalently,

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}.$$

Thus the least-squares solutions are exactly the solutions of the normal equations

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

Why is this true?

Every candidate fitted vector $A\mathbf{x}$ lies in $\text{col}(A)$. By the Projection theorem, $A\hat{\mathbf{x}}$ is closest to $\mathbf{b}$ exactly when

$$\mathbf{b} - A\hat{\mathbf{x}} \in \text{col}(A)^\perp.$$

By the fact above, this is equivalent to

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}.$$

Expanding and rearranging gives

$$A^T \mathbf{b} - A^T A \hat{\mathbf{x}} = \mathbf{0},$$

or

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

The fitted vector $\hat{\mathbf{b}}$ is unique because orthogonal projection onto a subspace is unique. The coefficient vector $\hat{\mathbf{x}}$ need not be unique. If $\mathbf{z} \in \text{null}(A)$, then

$$A(\hat{\mathbf{x}} + \mathbf{z}) = A\hat{\mathbf{x}}.$$

When the columns of A are linearly independent, $\text{null}(A) = \{\mathbf{0}\}$, so the least-squares coefficient vector is unique.

Activity: Independent columns and $A^T A$ (U2-LO5)

Let A be an $m \times n$ matrix. Use the identity

$$\mathbf{x}^T A^T A \mathbf{x} = \|A\mathbf{x}\|^2$$

to explain why $A^T A$ is invertible exactly when the columns of A are linearly independent.

1. Suppose

$$A^T A \mathbf{x} = \mathbf{0}.$$

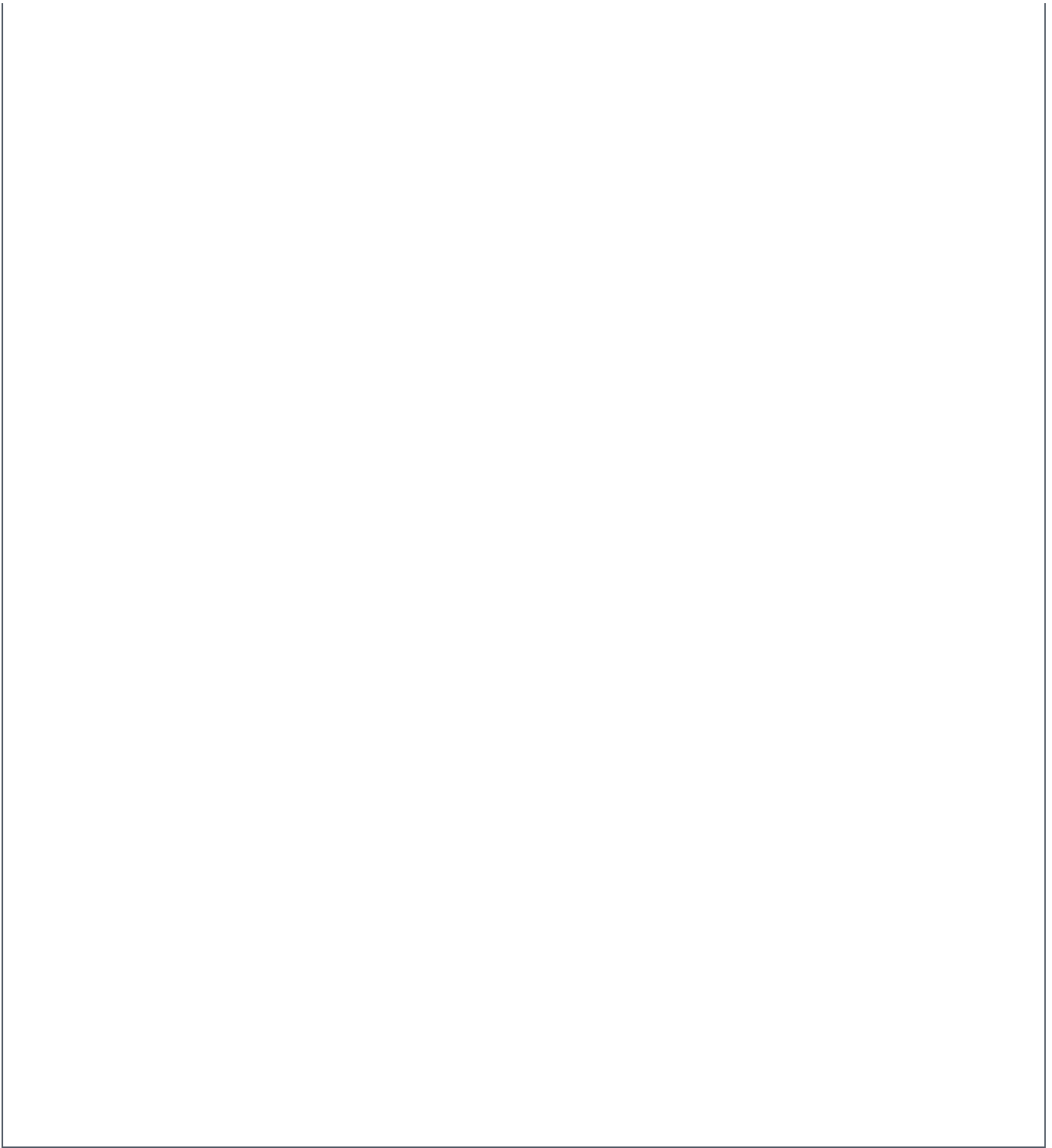
What does the displayed identity say about $\|A\mathbf{x}\|^2$?

2. If the columns of A are linearly independent, what must $\mathbf{x}$ be?
3. Conversely, suppose $A^T A$ is invertible and

$$A\mathbf{x} = \mathbf{0}.$$

What equation results after multiplying by A^T ?

4. Explain why the two conditions are equivalent.



Activity: Unit 2 target revisited (U4-LO4, U4-LO5)

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Recall why $A\mathbf{x} = \mathbf{b}$ has no exact solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations for $\hat{\mathbf{x}}$.
4. Compute the fitted vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}$$

and the residual

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

5. Check that

$$A^T \mathbf{r} = \mathbf{0}.$$

6. Compute $\|\mathbf{r}\|^2$.
7. Which vector is the closest reachable output: $\hat{\mathbf{x}}$ or $\hat{\mathbf{b}}$?
8. The residual is not zero. Explain why this does not mean least squares failed.

Suppose data are given as pairs

$$(t_1, y_1), \dots, (t_m, y_m).$$

A line model predicts

$$\hat{y} = c_0 + c_1 t.$$

The coefficient c_0 is the intercept, and c_1 is the slope. At input t_i , the predicted value is

$$\hat{y}_i = c_0 + c_1 t_i.$$

The residual for the i -th data point is

$$r_i = y_i - \hat{y}_i.$$

Linear regression chooses c_0 and c_1 to make

$$r_1^2 + \dots + r_m^2$$

as small as possible.

Linear regression is the same closest-reachable-output problem as “Unit 2 target revisited (U4-LO4, U4-LO5),” but the columns of the matrix now represent model features.

Activity: A line model as a matrix product (U4-LO4)

Consider the data points

$$(0, 1), \quad (1, 2), \quad (2, 2),$$

and the line model

$$\hat{y} = c_0 + c_1 t.$$

1. Write the three predicted values in terms of c_0 and c_1 .
2. Define

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Build a matrix X such that $X\mathbf{c}$ is the vector of predicted values.

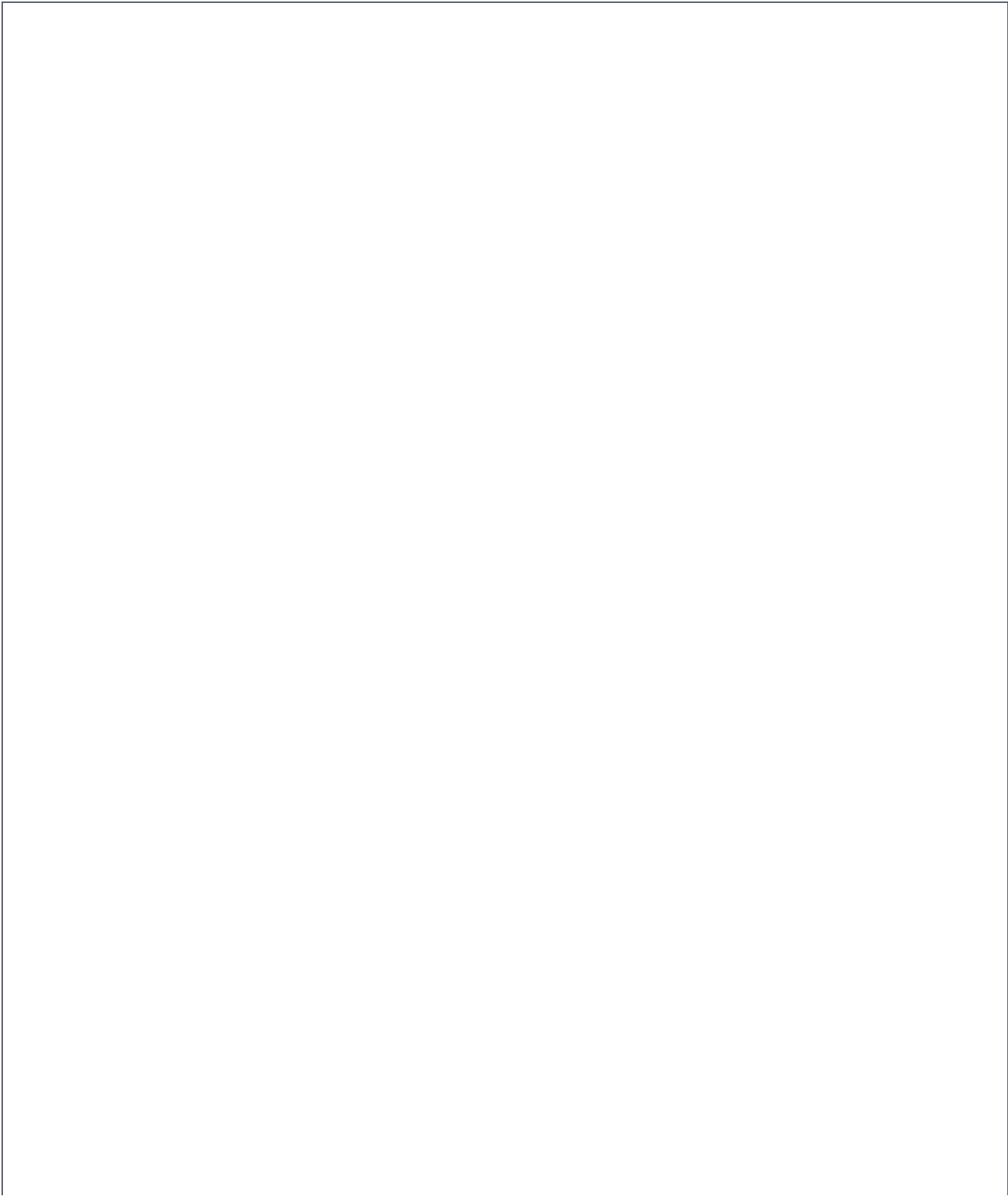
3. What does the first column of X represent?
4. What does the second column of X represent?
5. Write the residual vector

$$\mathbf{r} = \mathbf{y} - X\mathbf{c}.$$

6. Explain why

$$\|\mathbf{r}\|^2 = \sum_{i=1}^3 (y_i - \hat{y}_i)^2$$

is the sum of the squared vertical errors in the data plot.



For general data (t_i, y_i) , define the design matrix

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}.$$

Then $X\mathbf{c}$ is the vector of predicted values. Linear regression is the least-squares problem

$$\min_{\mathbf{c}} \|X\mathbf{c} - \mathbf{y}\|^2.$$

The matrix X is called the design matrix. Each column represents one feature used by the model.

Warning: Two different geometric pictures

The projection occurs in $\mathbb{R}^m$:

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}} = \text{proj}_{\text{col}(X)}(\mathbf{y}).$$

In the usual t - y data plot, the residuals are vertical differences

$$r_i = y_i - \hat{y}_i.$$

They are not generally perpendicular distances from the plotted points to the fitted line.

Activity: Fit a line to three data points (U4-LO4, U4-LO5)

Continue with the design matrix and data vector from “A line model as a matrix product (U4-LO4).”

1. Use “Least squares and the normal equations” to form

$$X^T X \hat{\mathbf{c}} = X^T \mathbf{y}.$$

2. Solve for

$$\hat{\mathbf{c}} = \begin{bmatrix} \hat{c}_0 \\ \hat{c}_1 \end{bmatrix}.$$

3. Write the fitted line.
4. Compute the fitted-value vector

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}}$$

and the residual vector

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}.$$

Activity: Fit a line to three data points (U4-LO4, U4-LO5) (continued)

5. Check that

$$X^T \mathbf{r} = \mathbf{0}.$$

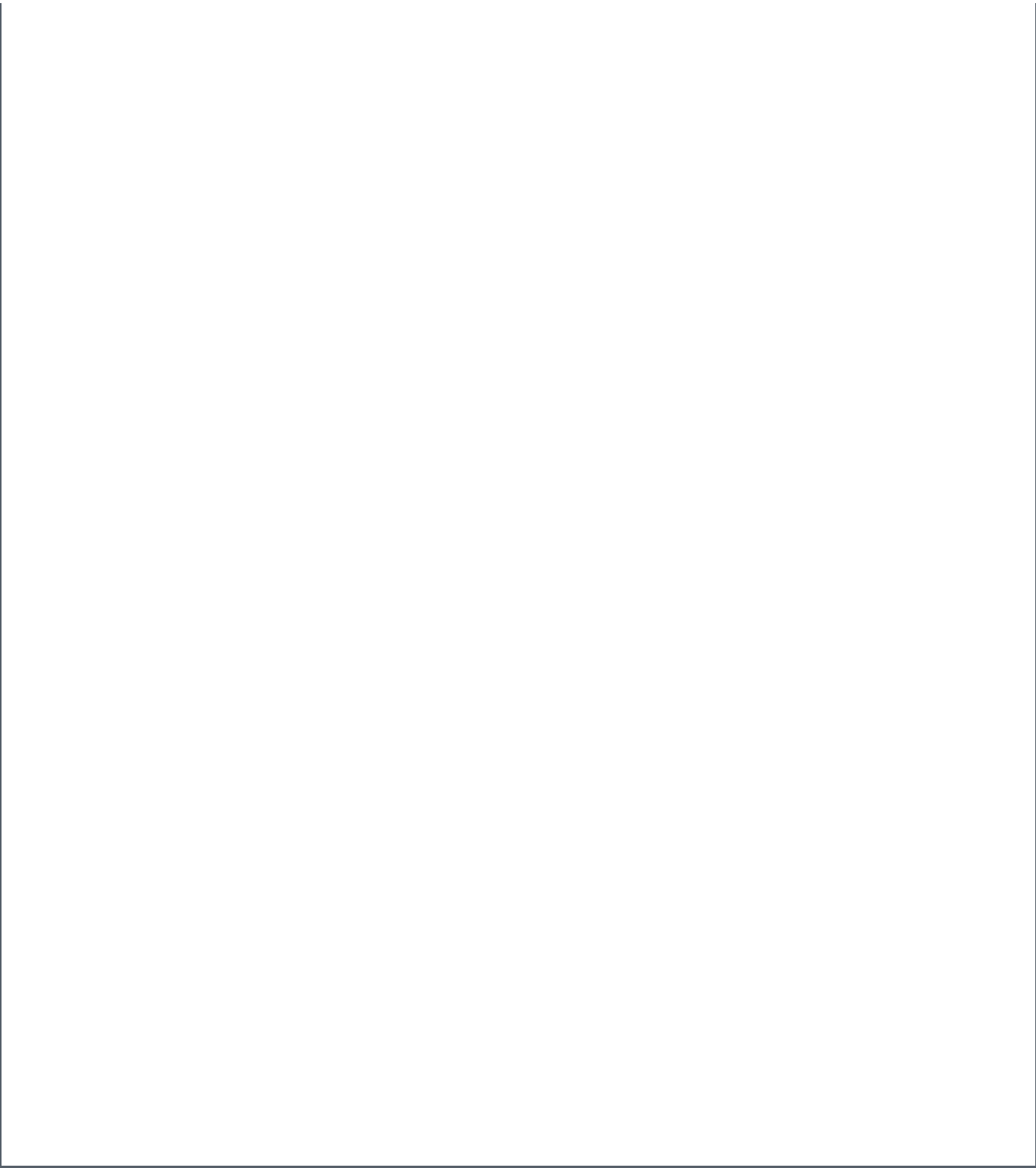
6. Write the two scalar equations contained in $X^T \mathbf{r} = \mathbf{0}$.

7. Compute

$$\|\mathbf{r}\|^2.$$

8. Which object is the fitted vector: $\hat{\mathbf{c}}$ or $\hat{\mathbf{y}}$?

9. In the t - y plot, are the residual segments vertical or perpendicular to the fitted line?



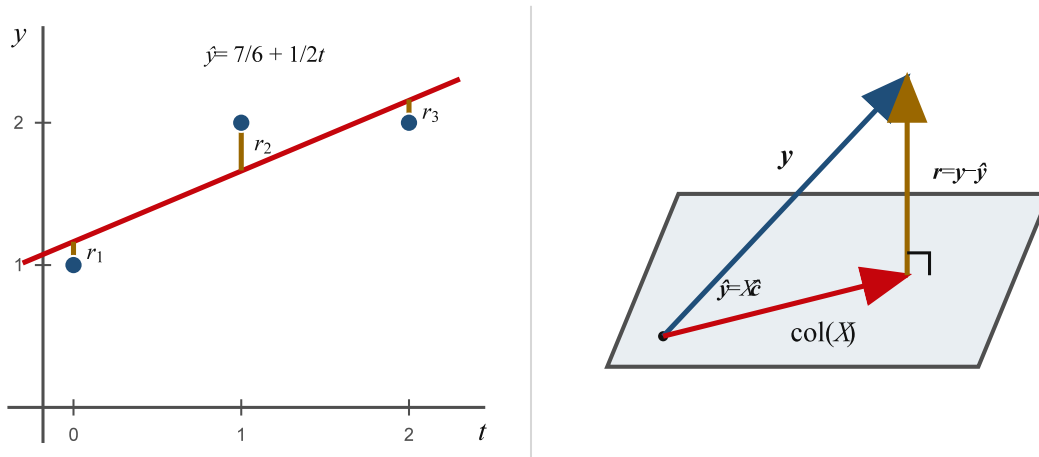


Figure. Two panels show the same linear regression. The left panel contains three data points, a fitted line, and vertical residual segments. The right panel shows the data vector y projected onto the column space of X , with a perpendicular residual r .

The two panels in the figure show different spaces: the left panel is the t - y data plot, while the right panel is the output space $\mathbb{R}^3$.

Activity: Reading regression code (U4-LO4, U4-LO7)

The following code repeats the line fit from “Fit a line to three data points (U4-LO4, U4-LO5).”

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

c = np.linalg.lstsq(X, y, rcond=None)[0]
yhat = X @ c
r = y - yhat

X, c, yhat, r, X.T @ r, np.linalg.norm(r)**2
```

1. What array is produced by `np.ones_like(t)`?

Activity: Reading regression code (U4-LO4, U4-LO7) (continued)

2. What does `np.column_stack([np.ones_like(t), t])` construct?
3. What is the shape of `X`?
4. What does `[0]` select from the value returned by `np.linalg.lstsq`?
5. Why does `c` have two entries?
6. What mathematical objects are stored in `yhat` and `r`?
7. Why is the residual check `X.T @ r` rather than `X @ r`?
8. What should `X.T @ r` be close to?
9. Does a nonzero residual vector mean the code failed?
10. What quantity is computed by `np.linalg.norm(r)**2`?

The normal equations describe the geometry of least squares. The next section uses Gram-Schmidt algorithm and QR factorization and least

squares to compute the same solution in a different form.

Explore least squares interactively with the [Residual Squares mini-game](#).